

IP ADDRESSING

All the computers of the world on the Internet network communicate with each other with underground or underwater cables or wirelessly. If I want to download a file from the internet or load a web page or literally do anything related to the internet, my computer must have an address so that other computers can find and locate mine in order to deliver that particular file or webpage that I am requesting. In technical terms, that address is called **IP Address or Internet Protocol Address**.

Working of IP addresses

The working of IP addresses is similar to other languages. It can also use some set of rules to send information. Using these protocols we can easily send, and receive data or files to the connected devices. There are several steps behind the scenes. Let us look at them

- Your device directly requests your Internet Service Provider which then grants your device access to the web.
- And an IP Address is assigned to your device from the given range available.
- Your internet activity goes through your service provider, and they route it back to you, using your IP address.
- Your IP address can change. For example, turning your router on or off can change your IP Address.
- When you are out from your home location your home IP address doesn't accompany you. It changes as you change the network of your device.

Types of IP Address

IP Address is of two types:

1. IPv4: Internet Protocol version 4. It consists of 4 numbers separated by the dots. Each number can be from 0-255 in decimal numbers. But computers do not understand decimal numbers, they instead change them to binary numbers which are only 0 and 1. Therefore, in binary, this (0-255) range can be written as (00000000 – 11111111). Since each number N can be represented by a group of 8-digit binary digits. So, a whole IPv4 binary address can be represented by 32-bits of binary digits. In IPv4, a unique sequence of bits is assigned to a computer, so a total of (2^{32}) devices approximately = 4,294,967,296 can be assigned with IPv4.

IPv4 can be written as:

189.123.123.90

Classes of IPv4 Address: There are around 4.3 billion IPv4 addresses and managing all those addresses without any scheme is next to impossible. Let's understand it with a simple example. If you have to find a word from a language dictionary, how long will it take? Usually, you will take less than 5 minutes to find that word. You are able to do this because words in the dictionary are organized in alphabetical order. If you have to find out the same word from a dictionary that doesn't use any sequence or order to organize the words, it will take an eternity to find the word. If a dictionary with one billion words without order can be so disastrous, then you can imagine the pain behind finding an address from 4.3 billion addresses. For easier management and assignment IP addresses are organized in numeric order and divided into the following 5 classes :

IP Class	Address Range	Maximum number of networks
Class A	1-126	126 (2^7-2)
Class B	128-191	16384
Class C	192-223	2097152

IP Class	Address Range	Maximum number of networks
Class D	224-239	Reserve for multitasking
Class E	240-254	Reserved for Research and development

The 0.0.0.0 is a [Non-routable address](#) is that indicates an invalid, or inapplicable end-user address.

A [loopback address](#) is a distinct reserved IP address range that starts from 127.0.0.0 ends at 127.255.255.255 though 127.255.255.255 is the broadcast address for 127.0.0.0/8. The loopback addresses are built into the IP domain system, enabling devices to transmit and receive the data packets. The loopback address 127.0.0.1 is generally known as localhost.

2. IPv6: But, there is a problem with the IPv4 address. With IPv4, we can connect only the above number of 4 billion devices uniquely, and apparently, there are much more devices in the world to be connected to the internet. So, gradually we are making our way to **IPv6 Address** which is a 128-bit IP address. In human-friendly form, IPv6 is written as a group of 8 hexadecimal numbers separated with colons(:). But in the computer-friendly form, it can be written as 128 bits of 0s and 1s. Since, a unique sequence of binary digits is given to computers, smartphones, and other devices to be connected to the internet. So, via IPv6 a total of (2^{128}) devices can be assigned with unique addresses which are actually more than enough for upcoming future generations.

IPv6 can be written as:

2011:0bd9:75c5:0000:0000:6b3e:0170:8394

Classification of IP Address

An IP address is classified into the following types:

1. Public IP Address: This address is available publicly and it is assigned by your network provider to your router, which further divides it to your devices. Public IP Addresses are of two types,

- **Dynamic IP Address:** When you connect a smartphone or computer to the internet, your Internet Service Provider provides you an IP Address from the range of available IP Addresses. Now, your device has an IP Address and you can simply connect your device to the Internet and send and receive data to and from your device. The very next time when you try to connect to the internet with the same device, your provider provides you with different IP Addresses to the same device and also from the same available range. Since IP Address keeps on changing every time when you connect to the internet, it is called a Dynamic IP Address.
- **Static IP Address:** Static address never changes. They serve as a permanent internet address. These are used by DNS servers. What are DNS servers? Actually, these are computers that help you to open a website on your computer. Static IP Address provides information such as device is located on which continent, which country, which city, and which Internet Service Provider provides internet connection to that particular device. Once, we know who is the ISP, we can trace the location of the device connected to the internet. Static IP Addresses provide less security than Dynamic IP Addresses because they are easier to track.

2. Private IP Address: This is an internal address of your device which are not routed to the internet and no exchange of data can take place between a private address and the internet.

3. Shared IP addresses: Many websites use shared IP addresses where the traffic is not huge and very much controllable, they decide to rent it to other similar websites so to make it cost-friendly. Several companies and email sending servers use the same IP address (within a single mail server) to cut down the cost so that they could save for the time the server is idle.

4. Dedicated IP addresses: A dedicated IP Address is an address used by a single company or an individual which gives them certain benefits using a private Secure Sockets Layer (SSL) certificate which is not in the case of a shared IP address. It allows to access the website or log in via File Transfer Protocol (FTP) by IP address instead of its domain name. It increases the performance of the website when the traffic is high. It also protects from a shared IP address that is black-listed due to spam.

Lookup IP addresses

To know your public IP, you can simply search “What is my IP?” on google. Other websites will show you equivalent information: they will see your public IP address because, by visiting the location, your router has made an invitation/request and thus revealed the information. the location IP location goes further by showing the name of your Internet Service Provider and your current city.

Finding your device’s private IP Address depends on the OS or platform you are using.

- **On Windows:** Click Start and type “cmd” in the search box and run the command prompt. In the black command prompt dialog box type “ipconfig” and press enter. You will be able to see your IP Address there.
- **On Mac:** Go to system preferences and select Network, you will be able to see the information regarding your network which includes your IP Address.

IP address security threats

Each IP address is associated with virtual ports in a computer that acts as a doorway that allows web applications or websites to send and receive data or information on your device. If after the connection is terminated the ports remain open somehow, might allow hackers to get into your device. Once, a hacker gets access to your device remotely through various tools and viruses, they would be able to access all your stored files and data and your computer hardware as well, which includes your webcam, mic, speaker, and all your browsing history, your emails and saved passwords. These are some serious threats from which we need to be extra careful.

Various online activities can reveal your IP address from playing games or accepting bad cookies from a trap website or commenting on a website or forum. Once, they have your IP, there are websites that help them get a decent idea of your location. They can further use social media websites to track your online presence and cross verify everything that they got from these sites and use your information for their benefits or can sell these data collected on the dark web which can further exploit you.

The worst which I have seen in my friend’s pc got infected while he was installing an application that he downloaded from a pirated website. The moment he hit install, a number of command prompt boxes started appearing, tens of commands started running and after a while, it was back to normal. Some malware was installed in the process. After a few days, someone was trying to log in to his social media account and other accounts using his computer as a host pc (his own IP address) but his computer was idle. The hacker was using his pc and his network, i.e., his IP address to do some serious stuff. He formatted his computer then and there, secured all his emails and other accounts, and changed all the passwords and all the security measures that had to be taken.

Cybercriminals use different techniques to get hands-on with your IP address and know your location, get into your network and hack into your computers. For instance, they will find you through Skype which uses IP addresses to speak. If you are using these apps, it’s important to notice that your IP address might be vulnerable. Attackers can use the various tools, where they will find your IP address. Some of the threats are: Online stalking, downloading illegal content using your IP address, tracking your location, directly attacking your network, and hacking into your device.

Protect and hide IP address

To secure and hide your IP address from unwanted people always remember the following points:

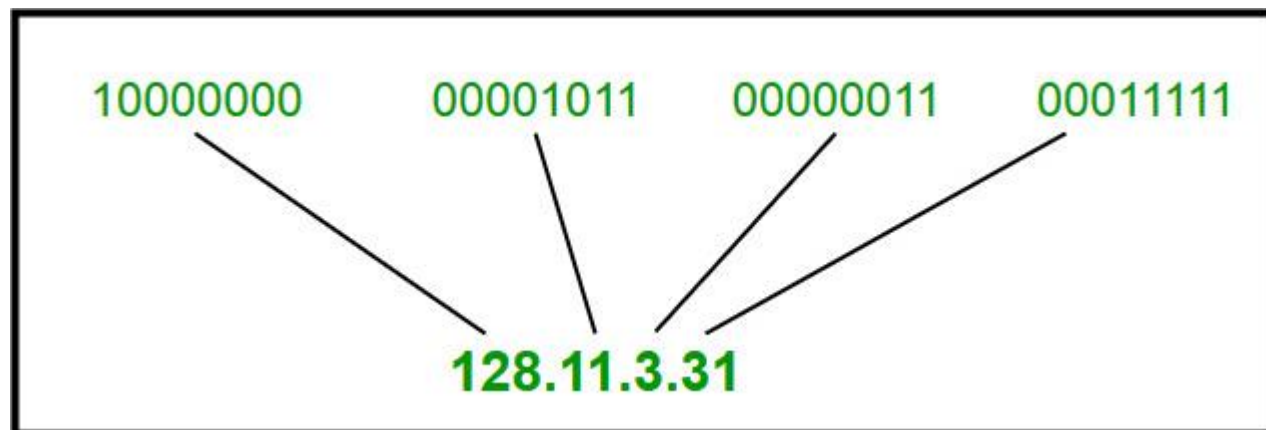
- Use a proxy server.
- Use a virtual private network (VPN) when using public Wi-Fi, you are traveling, working remotely, or just want some privacy.
- Change privacy settings on instant messaging applications.
- Create unique passwords.
- Beware of phishing emails and malicious content.
- Use a good and paid antivirus application and keep it up to date.
- When you are using public wifi in a cafe or station or anywhere, you must hide your IP address by using VPN. Getting your IP from public wifi is just a cakewalk for these hackers and they are very good at stealing all your information while using your computer's address. There are different phishing techniques in which they email you, call you, and SMS you about giving vital information about you. They give links to vicious websites which are pre-rigged. The moment you open these websites, they steal all your device's information revealing all the information about you and your device which are to be kept private. These leaks help the hackers to exploit your device and install or download some spyware and malware on your device. But using a good anti-virus gives you web security as well, which will prevent those websites to launch and warn you about the information being passed to these websites.
- It is also not recommended to use torrent or pirated websites which are a threat to your online identity and can compromise your device or emails or any other information about you.

Introduction of Classful IP Addressing

An IP address is an address having information about how to reach a specific host, especially outside the [LAN](#). An [IP address](#) is a 32-bit unique address having an address space of 2^{32} .

Generally, there are two notations in which the IP address is written, dotted decimal notation and hexadecimal notation.

Dotted Decimal Notation



Dotted Decimal Notation

Hexadecimal Notation



Some points to be noted about dotted decimal notation:

1. The value of any segment (byte) is between 0 and 255 (both included).
2. No zeroes are preceding the value in any segment (054 is wrong, 54 is correct).

Classful Addressing

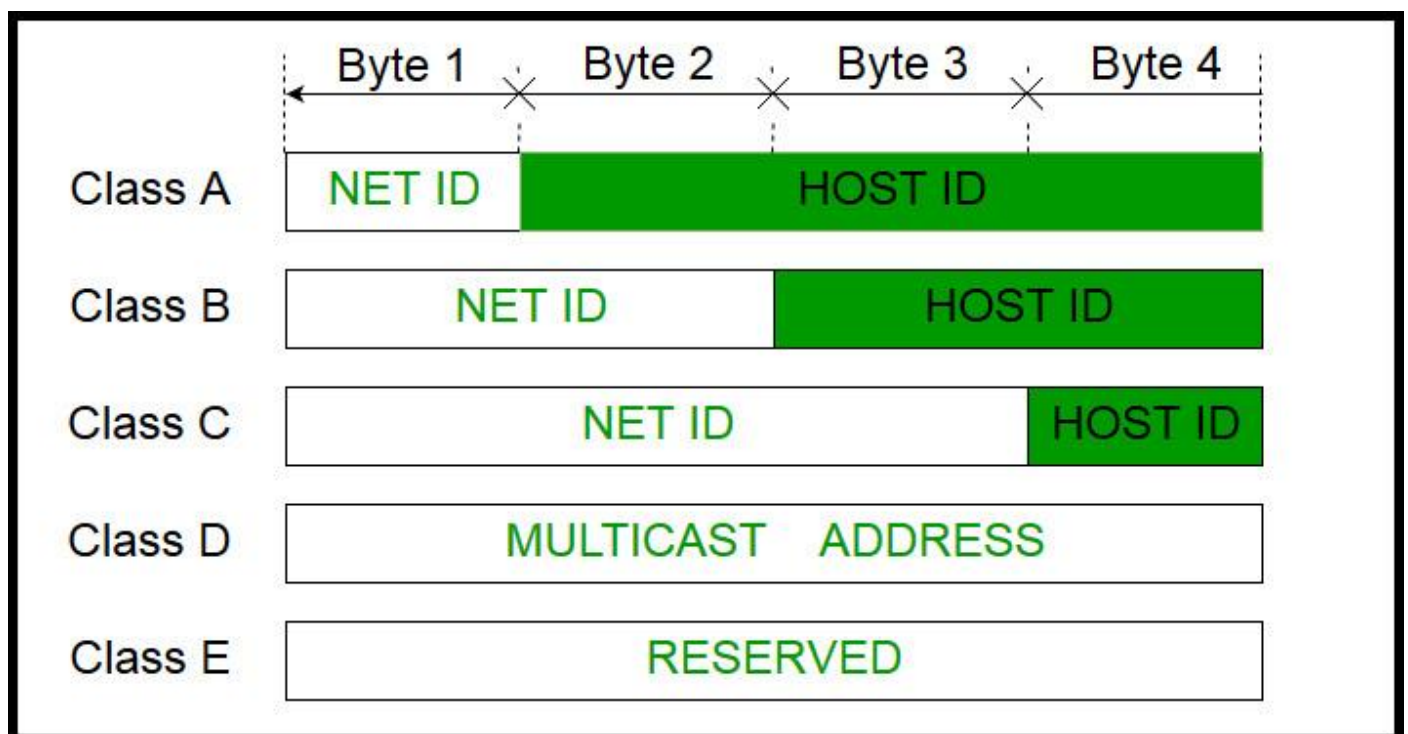
The 32-bit IP address is divided into five sub-classes. These are:

- Class A
- Class B
- Class C
- Class D
- Class E

Each of these classes has a valid range of IP addresses. Classes D and E are reserved for multicast and experimental purposes respectively. The order of bits in the first octet determines the classes of the IP address. The IPv4 address is divided into two parts:

- Network ID
- Host ID

The class of IP address is used to determine the bits used for network ID and host ID and the number of total networks and hosts possible in that particular class. Each ISP or network administrator assigns an IP address to each device that is connected to its network.



Classful Addressing

Note:

1. IP addresses are globally managed by Internet Assigned Numbers Authority(IANA) and regional Internet registries(RIR).
2. While finding the total number of host IP addresses, 2 IP addresses are not counted and are therefore, decreased from the total count because the first IP address of any network is the network number and whereas the last IP address is reserved for broadcast IP.

Class A

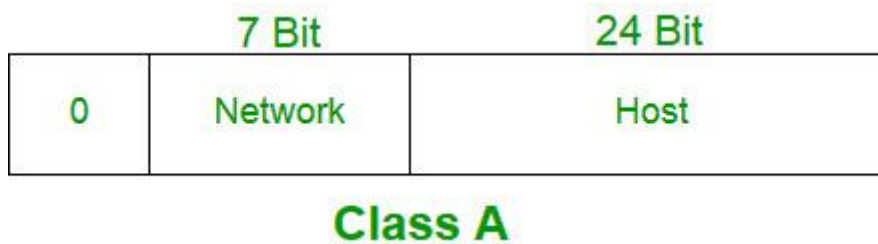
IP addresses belonging to class A are assigned to the networks that contain a large number of hosts.

- The network ID is 8 bits long.
- The host ID is 24 bits long.

The higher-order bit of the first octet in class A is always set to 0. The remaining 7 bits in the first octet are used to determine network ID. The 24 bits of host ID are used to determine the host in any network. The default subnet mask for Class A is 255.x.x.x. Therefore, class A has a total of:

- $2^{24} - 2 = 16,777,214$ host ID

IP addresses belonging to class A ranges from 1.x.x.x – 126.x.x.x



Class A

Class B

IP address belonging to class B is assigned to networks that range from medium-sized to large-sized networks.

- The network ID is 16 bits long.
- The host ID is 16 bits long.

The higher-order bits of the first octet of IP addresses of class B are always set to 10. The remaining 14 bits are used to determine the network ID. The 16 bits of host ID are used to determine the host in any network. The default subnet mask for class B is 255.255.x.x. Class B has a total of:

- $2^{14} = 16384$ network address
- $2^{16} - 2 = 65534$ host address

IP addresses belonging to class B ranges from 128.0.x.x – 191.255.x.x.



Class B

Class B

Class C

IP addresses belonging to class C are assigned to small-sized networks.

- The network ID is 24 bits long.
- The host ID is 8 bits long.

The higher-order bits of the first octet of IP addresses of class C is always set to 110. The remaining 21 bits are used to determine the network ID. The 8 bits of host ID are used to determine the host in any network. The default subnet mask for class C is 255.255.255.x. Class C has a total of:

- $2^{21} = 2097152$ network address
- $2^8 - 2 = 254$ host address

IP addresses belonging to class C range from 192.0.0.x – 223.255.255.x.



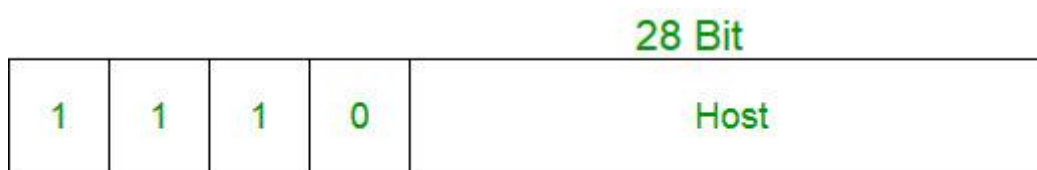
Class C

Class C

Class D

IP address belonging to class D is reserved for multi-casting. The higher-order bits of the first octet of IP addresses belonging to class D is always set to 1110. The remaining bits are for the address that interested hosts recognize.

Class D does not possess any subnet mask. IP addresses belonging to class D range from 224.0.0.0 – 239.255.255.255.

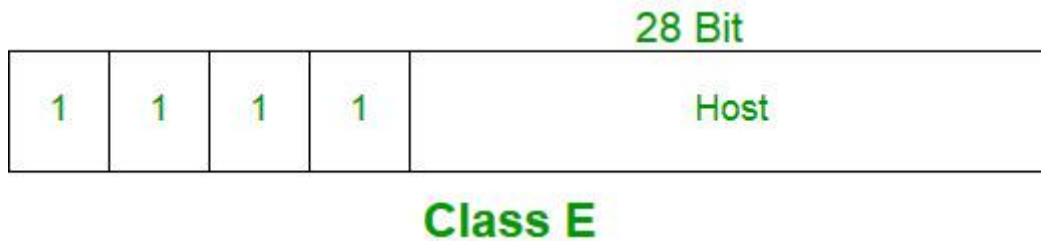


Class D

Class D

Class E

IP addresses belonging to class E are reserved for experimental and research purposes. IP addresses of class E range from 240.0.0.0 – 255.255.255.254. This class doesn't have any subnet mask. The higher-order bits of the first octet of class E are always set to 1111.



Class E

Range of Special IP Addresses

169.254.0.0 – 169.254.0.16 : Link-local addresses

127.0.0.0 – 127.0.0.8 : Loop-back addresses

0.0.0.0 – 0.0.0.8: used to communicate within the current network.

Rules for Assigning Host ID

Host IDs are used to identify a host within a network. The host ID is assigned based on the following rules:

- Within any network, the host ID must be unique to that network.
- A host ID in which all bits are set to 0 cannot be assigned because this host ID is used to represent the network ID of the IP address.
- Host ID in which all bits are set to 1 cannot be assigned because this host ID is reserved as a broadcast address to send packets to all the hosts present on that particular network.

Rules for Assigning Network ID

Hosts that are located on the same physical network are identified by the network ID, as all host on the same physical network is assigned the same network ID. The network ID is assigned based on the following rules:

- The network ID cannot start with 127 because 127 belongs to the class A address and is reserved for internal loopback functions.
- All bits of network ID set to 1 are reserved for use as an IP broadcast address and therefore, cannot be used.
- All bits of network ID set to 0 are used to denote a specific host on the local network and are not routed and therefore, aren't used.

Summary of Classful Addressing

CLASS	LEADING BITS	NET ID BITS	HOST ID BITS	NO. OF NETWORKS	ADDRESSES PER NETWORK	START ADDRESS	END ADDRESS
CLASS A	0	8	24	2^7 (128)	2^{24} (16,777,216)	0.0.0.0	127.255.255.255
CLASS B	10	16	16	2^{14} (16,384)	2^{16} (65,536)	128.0.0.0	191.255.255.255
CLASS C	110	24	8	2^{21} (2,097,152)	2^8 (256)	192.0.0.0	223.255.255.255
CLASS D	1110	NOT DEFINED	NOT DEFINED	NOT DEFINED	NOT DEFINED	224.0.0.0	239.255.255.255
CLASS E	1111	NOT DEFINED	NOT DEFINED	NOT DEFINED	NOT DEFINED	240.0.0.0	255.255.255.255

Problems with Classful Addressing

The problem with this classful addressing method is that millions of class A addresses are wasted, many of the class B addresses are wasted, whereas, the number of addresses available in class C is so small that it cannot cater to the needs of organizations. Class D addresses are used for multicast routing and are therefore available as a single block only. Class E addresses are reserved.

Since there are these problems, Classful networking was replaced by Classless Inter-Domain Routing (CIDR) in 1993. We will be discussing Classless addressing in the next post.

- The network ID is 24 bits long.
- The host ID is 8 bits long.
- $2^{21} = 2097152$ network address
- $2^8 - 2 = 254$ host address
- Within any network, the host ID must be unique to that network.
- Host ID in which all bits are set to 0 cannot be assigned because this host ID is used to represent the network ID of the IP address.
- Host ID in which all bits are set to 1 cannot be assigned because this host ID is reserved as a broadcast address to send packets to all the hosts present on that particular network.
- The network ID cannot start with 127 because 127 belongs to the class A address and is reserved for internal loopback functions.
- All bits of network ID set to 1 are reserved for use as an IP broadcast address and therefore, cannot be used.
- All bits of network ID set to 0 are used to denote a specific host on the local network and are not routed and therefore, aren't used.

Classless Addressing in IP Addressing

CIDR (Classless Inter-Domain Routing or supernetting) is a method of assigning [IP](#) addresses that improves the efficiency of address distribution and replaces the previous system based on Class A, Class B and Class C networks.

The Network address identifies a network on the internet. Using this, we can find a range of addresses in the network and total possible number of hosts in the network.

Mask is a 32-bit binary number that gives the network address in the address block when AND operation is bitwise applied on the mask and any IP address of the block.

The default masks in different classes are :

- Class A – 255.0.0.0
- Class B – 255.255.0.0
- Class C – 255.255.255.0

Classless Addressing

To reduce the wastage of IP addresses in a block, we use sub-netting. What we do is that we use host id bits as net id bits of a classful IP address. We give the IP address and define the number of bits for mask along with it (usually followed by a '/' symbol), like, 192.168.1.1/28. Here, subnet mask is found by putting the given number of bits out of 32 as 1, like, in the given address, we need to put 28 out of 32 bits as 1 and the rest as 0, and so, the subnet mask would be 255.255.255.240. A classless addressing system or classless interdomain routing (CIDR or supernetting) is the way to combine two or more class C networks to create a /23 or a /22 supernet. A classless addressing system or classless interdomain routing (CIDR) is an improved IP addressing system. In a classless addressing system the block of IP address is assigned dynamically based on specific rules.

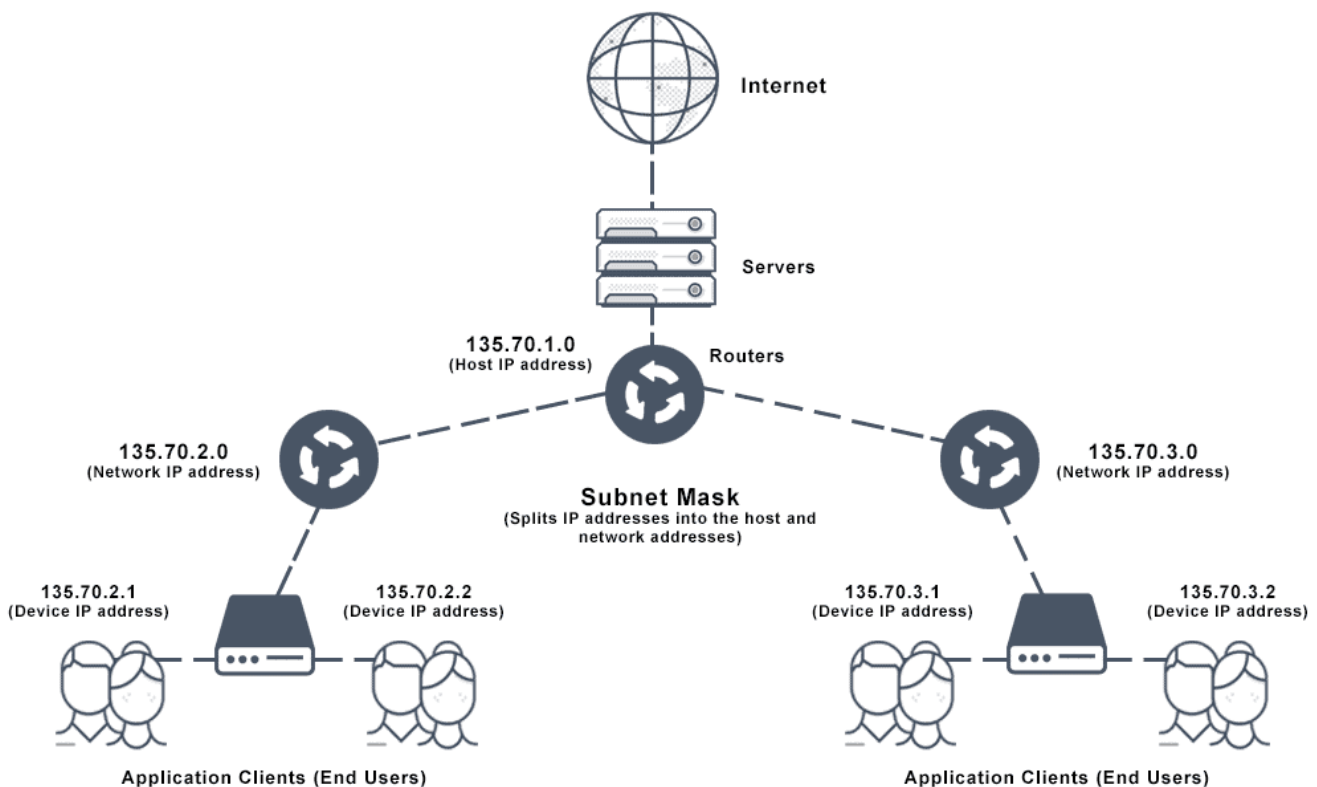
Some Values Calculated in Subnetting:

1. Number of subnets : $2^{(\text{Given bits for mask} - \text{No. of bits in default mask})}$
2. Subnet address : AND result of subnet mask and the given IP address
3. Broadcast address : By putting the host bits as 1 and retaining the network bits as in the IP address
4. Number of hosts per subnet : $2^{(32 - \text{Given bits for mask})} - 2$
5. First Host ID : Subnet address + 1 (adding one to the binary representation of the subnet address)

Subnet Mask-

Every device has an IP address with two pieces: the client or host address and the server or network address. IP addresses are either configured by a DHCP server or manually configured (static IP addresses). The subnet mask splits the IP address into the host and network addresses, thereby defining which part of the IP address belongs to the device and which part belongs to the network.

The device called a gateway or default gateway connects local devices to other networks. This means that when a local device wants to send information to a device at an IP address on another network, it first sends its packets to the gateway, which then forwards the data on to its destination outside of the local network.



What is Subnet Mask?

A subnet mask is a 32-bit number created by setting host bits to all 0s and setting network bits to all 1s. In this way, the subnet mask separates the IP address into the network and host addresses.

The “255” address is always assigned to a broadcast address, and the “0” address is always assigned to a network address. Neither can be assigned to hosts, as they are reserved for these special purposes.

The IP address, subnet mask and gateway or router comprise an underlying structure—the Internet Protocol—that most networks use to facilitate inter-device communication.

When organizations need additional subnetworking, subnetting divides the host element of the IP address further into a subnet. The goal of subnet masks are simply to enable the subnetting process. The phrase “mask” is applied because the subnet mask essentially uses its own 32-bit number to mask the IP address.

IP Address and Subnet Mask

A 32-bit IP address uniquely identifies a single device on an IP network. The 32 binary bits are divided into the host and network sections by the subnet mask but they are also broken into four 8-bit octets.

Because binary is challenging, we convert each octet so they are expressed in dot decimal. This results in the characteristic dotted decimal format for IP addresses—for example, 172.16.254.1. The range of values in decimal is 0 to 255 because that represents 00000000 to 11111111 in binary.

IP Address Classes and Subnet Masks

Since the internet must accommodate networks of all sizes, an addressing scheme for a range of networks exists based on how the octets in an IP address are broken down. You can determine based on the three high-order or left-most bits in any given IP address which of the five different classes of networks, A to E, the address falls within.

(Class D networks are reserved for multicasting, and Class E networks not used on the internet because they are reserved for research by the Internet Engineering Task Force IETF.)

A Class A subnet mask reflects the network portion in the first octet and leaves octets 2, 3, and 4 for the network manager to divide into hosts and subnets as needed. Class A is for networks with more than 65,536 hosts.

A Class B subnet mask claims the first two octets for the network, leaving the remaining part of the address, the 16 bits of octets 3 and 4, for the subnet and host part. Class B is for networks with 256 to 65,534 hosts.

In a Class C subnet mask, the network portion is the first three octets with the hosts and subnets in just the remaining 8 bits of octet 4. Class C is for smaller networks with fewer than 254 hosts.

Class A, B, and C networks have natural masks, or default subnet masks:

- Class A: 255.0.0.0
- Class B: 255.255.0.0
- Class C: 255.255.255.0

You can determine the number and type of IP addresses any given local network requires based on its default subnet mask.

An example of Class A IP address and subnet mask would be the Class A default submask of 255.0.0.0 and an IP address of 10.20.12.2.

How Does Subnetting Work?

Subnetting is the technique for logically partitioning a single physical network into multiple smaller sub-networks or subnets.

Subnetting enables an organization to conceal network complexity and reduce network traffic by adding subnets without a new network number. When a single network number must be used across many segments of a local area network (LAN), subnetting is essential.

The benefits of subnetting include:

- Reducing broadcast volume and thus network traffic
- Enabling work from home
- Allowing organizations to surpass LAN constraints such as maximum number of hosts

Network Addressing

The standard modern network prefix, used for both IPv6 and IPv4, is Classless Inter-Domain Routing (CIDR) notation. IPv4 addresses represented in CIDR notation are called network masks, and they specify the number of bits in the prefix to the address after a forward slash (/) separator. This is the sole standards-based format in IPv6 to denote routing or network prefixes.

To assign an IP address to a network interface since the advent of CIDR, there are two parameters: a subnet mask and the address. Subnetting increases routing complexity, because there must be a separate entry in each connected router's tables to represent each locally connected subnet.

What Does IP Mask Mean?

Typically, although the phrase "subnet mask" is preferred, you might use "IP/Mask" as a shorthand to define both the IP address and submask at once. In this situation, the IP address is followed by the number of bits in the mask. For example:

10.0.1.1/24

216.202.192.66/22

These are equivalent to

IP address: 10.0.1.1 with subnet mask of 255.255.255.0

IP address: 216.202.196.66 with a subnet mask example of 255.255.252.0

However, you do not mask the IP address, you mask the subnet.

For more on the actual implementation of load balancing, security applications and web application firewalls check out our [Application Delivery How-To Videos](#).